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SUPERSEDING
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MILITARY SPECIFICATION

TRANSMITTER, THRUST, SINGLE SYNCHRO STYLE, PRESSURE RATIO, TYPE MK-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pressure ratio, synchro style thrust transmitter, designated type MK-2.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-E-1	Electron Tube, General Specification For
MIL-P-116	Preservation, Methods Of
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion On
MIL-C-5015	Connector, Electric, AN Type
MIL-T-5350	Transmitters, Synchro, Aircraft, General Specification For
MIL-I-25859	Indicator, Thrust AEU-2/A, Synchro

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization Of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing Of
MIL-STD-810	Environmental Test Methods
MS33649	Bosses, Fluid Connection-Internal Straight Thread
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 First article test. The item to be produced under this specification requires First Article Test (see 4.4).

3.2 General specification. The requirements of MIL-T-5350 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

*3.2.1 Exception to the requirements of MIL-T-5350. The seasoning and acceleration tests of MIL-T-5350 shall not be required by this specification. Also, the extreme high temperature test and the electromagnetic interference test shall not be required.

3.3 Design and construction. The transmitter shall be designed to sense compressor inlet total pressure (P_{T2}), Turbine outlet total pressure (P_{T7}), to divide P_{T2} by P_{T7} and provide a synchro signal for indication of this ratio.

3.3.1 Stops. The mechanism of the transmitter shall contain suitable stops which allow an indication beyond specified range (both minimum and maximum) of 0.1 to 0.2 pressure ratio.

*3.3.2 Pressure ranges. The compressor inlet pressure (P_{T2}) range shall be 3 to 53 inches Hg absolute. The Turbine discharge total pressure (P_{T7}) range shall be 3 to 106 inches Hg absolute. The differential pressure (P_{T7}-P_{T2}) range shall be

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3.3.3 Compatibility. The calibration of the transmitter shall be such that it will satisfactorily operate with single synchro style indicators in accordance with MIL-I-25859.

3.3.4 Case. The transmitter case shall be sealed (except for the pressure connections). Sealing may be accomplished hermetically or with "O" ring or other type gaskets.

3.3.4.1 Safety wiring. External body screws shall be secured with safety wire or suitable lock washers. Lock washers shall be approved by the qualifying activity. Internal screws (where applicable) shall be safetied with either lock wire or washers. The method selected shall be approved by the qualifying activity.

3.3.5 Damping. The transmitter shall be provided with suitable damping.

3.3.6 Dimensions. The outline dimensions of the transmitter shall not exceed those referenced in Figure 1.

3.3.7 Mounting. Mounting bosses and location thereof shall be as shown in Figure 1. A mounting rack shall be used.

3.3.8 Fittings. The P_{T2} and P_{T7} pressure connecting fittings and location shall be as shown in Figure 1. Metal tubing shall be used to connect rack fittings to transmitter fittings, if required.

3.3.8.1 Caps. The pressure connections shall be supplied with caps to protect the threads and prevent the entrance of foreign material. Plugs or tape shall not be used.

3.3.9 Vacuum tubes. Vacuum tubes, if used, shall be of ruggedized construction conforming to MIL-E-1.

3.3.10 Transistors and diodes. Transistors and diodes shall be of the silicon type designed for continuous operation in a 121°C ambient temperature in accordance with MIL-STD-701.

3.3.11 Power supply. The transmitter shall be designed to operate from an aircraft 115V, 400 Hz power supply in accordance with MIL-STD-704. This voltage shall be suitably reduced within the transmitter to provide for operation of 26 Volt synchros as specified in MIL-T-5350.

3.3.11.1 Power consumption. Power consumption of the transmitter shall not exceed 90 va.

3.3.12 Electrical receptacle. The transmitter shall be provided with an MS33678-16S-1P receptacle conforming to MIL-C-5015.

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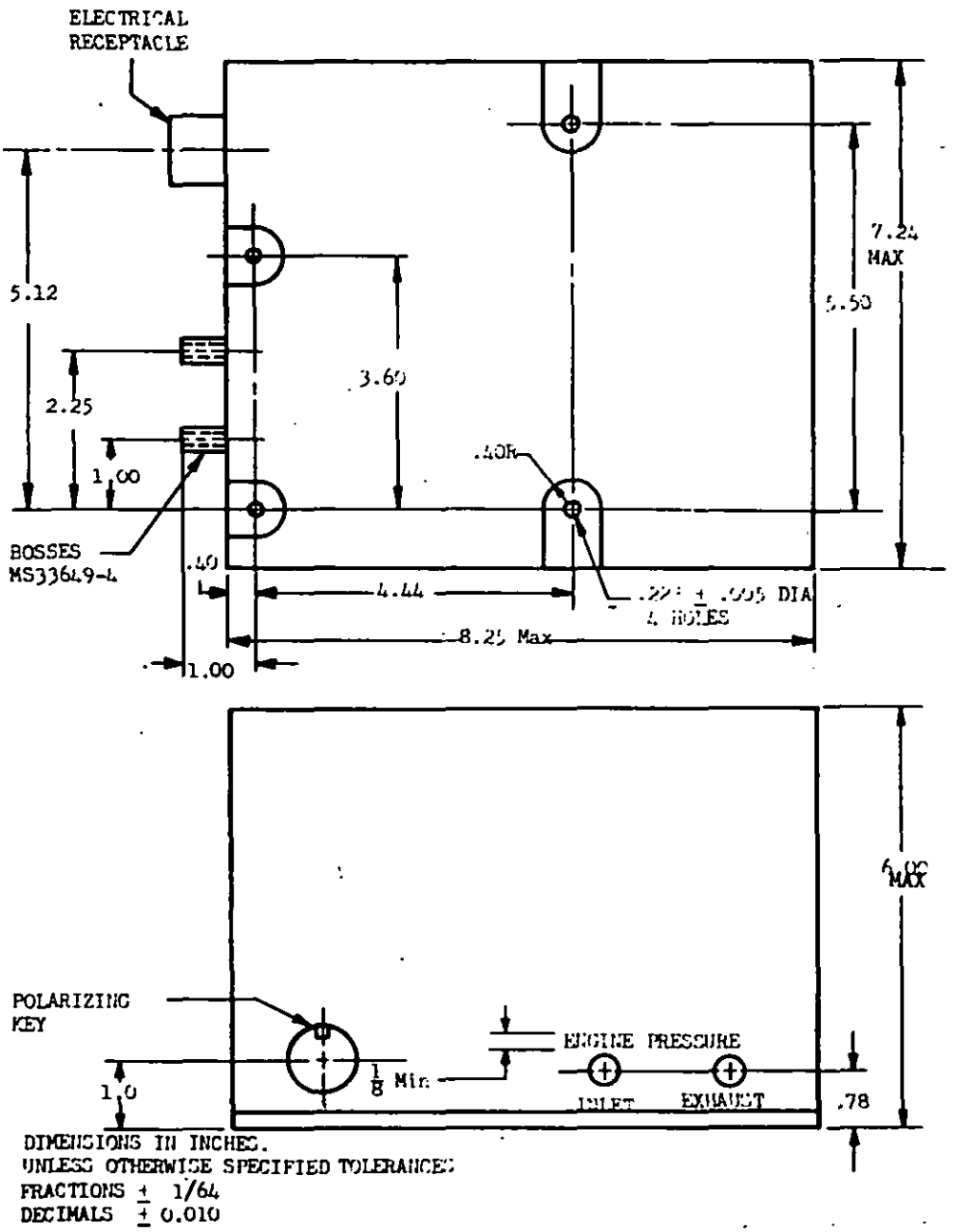


FIGURE 1 TRANSMITTER

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*3.3.13 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

*3.4 Performance. The transmitter shall be capable of meeting the requirements of MIL-T-5350 with the following exceptions:

a. Seasoning - no requirement

b. Endurance - 1000 hours (P_{T2} cycled between ambient and 9 ± .5 inches Hg at a rate of 5 ± 2 cycles per minute (cpm) for 500 hours and P_{T7} cycled between ambient and 100 ± 5 inches Hg at 5 ± 2 cpm for 500 hours).

c. Shock - Shock forces up to 15 g's

d. Acceleration - no requirement

e. Electro magnetic interference - no requirement

f. Extreme high temperature - no requirement

3.5 Weight. The weight of the transmitter shall not exceed 5.5 pounds.

3.6 Markings. The P_{T2} and P_{T7} pressure connections shall be permanently and legibly marked as shown in Figure 1. An external wiring diagram in accordance with Figure 2 shall be marked on the case by a decal, plate, or by stamping.

3.7 Magnesium alloys. Magnesium alloys may be used in the construction of the transmitter. Suitable protection against corrosion shall be provided in accordance with MIL-M-3171.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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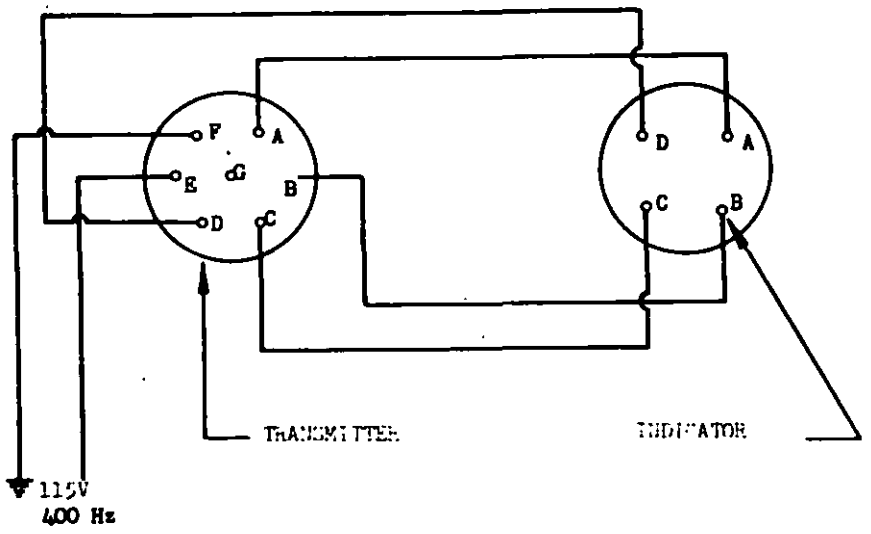


FIGURE 2. External wiring diagram.

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4.2 General specification. Sampling, inspection, and testing of the transmitter shall be in accordance with MIL-T-5350, except as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.6)

b. Quality Conformance Inspection (see 4.5)

*4.4 First article test. First article test shall consist of the test listed in paragraph 4.6.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of individual inspection and sampling plan inspections.

4.5.1 Individual inspection. In addition to the individual tests of MIL-T-5350, each transmitter shall be subjected to the leak test (4.6.1) of this specification. The position error test shall be conducted as specified under 4.6.4.

4.5.2 Sampling inspection.

4.5.2.1 Sampling Plan A. The scale error test shall be conducted as specified under 4.6.2.2.

4.5.2.2 Sampling Plan B. The seasoning and acceleration tests of MIL-T-5350 shall be omitted. The high temperature exposure, endurance, and shock tests shall be conducted as specified under 4.6.9, 4.6.10, and 4.6.11, respectively. In addition to the sampling plan B tests of MIL-T-5350, the overpressure test of 4.6.12 shall be performed.

*4.6 First article tests.

4.6.1 Leak. A pressure of 42 inches Hg absolute shall be applied to the P_{T2} port. The pressure line shall then be sealed upstream of the manometer and with the P_{T7} port open to ambient pressure, the P_{T2} pressure shall not decrease more than 0.05 inch Hg during 5 minutes. A pressure of 96 inches Hg absolute shall be applied to the P_{T7} port. The pressure line shall then be sealed upstream of the manometer and with the P_{T2} port held at approximately 35 inches Hg absolute, the P_{T7} pressure shall not decrease more than 0.05 inch during 5 minutes. The internal volume of the isolated part of the system (manometer wells, pressure lines, et cetera) shall not exceed 45 cubic inches.

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4.6.2 Scale error.

4.6.2.1 Room temperature. The transmitter shall be subjected to the scale error test specified in Table I at room temperature, only at the points marked with a double asterisk. During the test, the transmitter shall be vibrated at a frequency of 60 Hz with an amplitude not to exceed 0.002 inch.

4.6.2.2 High and low temperature. For sampling plan A, the scale error test shall be conducted at each of the following temperatures in the order listed. All the test points of Table I shall be conducted and the tolerances of Table I shall apply for scale error and friction. The transmitter shall be maintained at the appropriate temperature for 4 hours prior to the test. The temperature shall be held to a ± 2°C tolerance:

a. Room temperature

b. -55°C

c. -35°C

d. 0°C

e. +60°C

f. +93°C

g. Room temperature

4.6.2.3 Hysteresis. All test points are to be run with both increasing and decreasing pressure. The difference between the up scale and down scale readings shall not exceed one half the scale error tolerance at each test point. All test points shall fall within the allowable tolerance.

4.6.3 Friction error. The tolerance specified in Table I shall apply.

4.6.4 Position error. Position error shall be tested at positions of 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° around the major and minor axes. Position error will be tested only at those points marked with an asterisk in Table I and only at room temperature. The differenece between readings in normal operating position and any other position shall not exceed one half the scale error tolerance at each test point. This test may be conducted in conjunction with the scale error test.

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TABLE I
SCALE ERROR

P_{T2} Inches Hg Absolute	P_{T7} Inches Hg Absolute	P_{T7}/P_{T2}
2.00	2.4	1.20
2.00	4.8	2.40
2.00	5.6	2.80
2.00	6.8	3.40
4.00*	7.20	1.80
**4.00	8.00	2.00
4.00	8.80	2.20
**4.00	9.60	2.40
4.00	10.40	2.60
**6.50	11.70	1.80
**6.50*	13.00	2.00
**6.50	14.30	2.20
6.50	15.60	2.40
6.50	16.90	2.60
**10.50	18.90	1.80
10.50	21.00	2.00
**10.50*	23.10	2.20
10.50	25.20	2.40
**10.50	27.30	2.60
10.50	29.40	2.80
**25.00	40.00	1.60
25.00	45.00	1.80
25.00	50.00	2.00
**25.00	55.00	2.20
25.00	60.00	2.40
25.00	65.00	2.60
25.00	70.00	2.80
**25.00	75.00	3.00
30.00	48.00	1.60
**30.00	54.00	1.80
**30.00	60.00	2.00
**30.00	66.00	2.20
30.00	72.00	2.40
30.00	78.00	2.60
**30.00	84.00	2.80
40.00	48.00	1.20
40.00	56.00	1.40
40.00	64.00	1.60
**40.00	72.00	1.80
40.00	80.00	2.00
40.00	88.00	2.20
40.00	96.00	2.40

Room temperature scale error tolerance	0.020
High temperature tolerance	0.020
Low temperature scale error tolerance	0.030
Friction error tolerances	0.015 At all temperatures

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4.6.5 Dielectric strength. In the event the transmitter is internally grounded, or damage to the amplifier will result from this test, the dielectric strength test shall be conducted prior to assembly or prior to installation of the sensitive component.

4.6.6 Low temperature. Scale error and friction shall not exceed the tolerance specified in Table I.

4.6.7 High temperature. Scale error and friction shall not exceed the tolerance specified in Table I.

4.6.8 Vibration. Signal oscillation shall not exceed one half the scale error tolerance. Following the vibration test, the transmitter shall be subjected to the scale error test of 4.6.2.1.

*4.6.9 High temperature exposure. The test specified in MIL-T-5350 shall be conducted for 50 hours except that the temperature shall be +93°C ± 2°C.

4.6.10 Endurance. The transmitter shall be subjected to 1,000 hours of operation. Five hundred hours shall be accomplished while P_{T2} is varied from ambient to 9 ± 0.5 inches Hg absolute to ambient at a rate of 5 ± 2 cpm. The remaining 500 hours shall be accomplished while P_{T7} is varied from ambient to 100 ± 5 inches Hg absolute to ambient at a rate of 5 ± 2 cpm. The pressure sensing element not being cycled shall be vented to ambient. The transmitter shall then be subjected to the scale error test of 4.6.2.1.

4.6.11 Shock. The transmitter shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedure I, except as specified herein. The transmitter shall be subjected to three impact shocks of 15g's amplitude and 7 ± 2 milliseconds duration. One shock shall be applied along each of the three mutually perpendicular axis of the transmitter.

4.6.12 Overpressure. The transmitter shall be subjected to a P_{T2} pressure of 85 ± 1 inch Hg absolute and a P_{T7} pressure of 150 ± 1 inch Hg absolute. The resultant pressure ratio shall be accurate to ± 0.100 pressure ratio units. The room temperature scale error test shall be conducted with the errors not to exceed those for room temperature.

4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

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5. PREPARATION FOR DELIVERY

*5.1 Preservation and packaging. Preservation and packaging of transmitters shall be Level A or C as specified by the procuring activity (see paragraph 6.2). Reference MIL-STD-794 for definition of levels of preservation and packaging.

*5.1.2 Level A. Transmitters shall be preserved and packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Cleaning and drying shall be accomplished in accordance with applicable methods of MIL-P-116. Overbox completed packs in PPP-B-636 unit container. Apply sufficient cushioning material on all faces between primary and unit container of a type, density, and thickness to insure shock transmission to completed pack does not exceed the peak value in Gs established for item, when subjected to the gough handling drop tests of MIL-P-116.

*5.2 Packing. Packing shall be Level A, B or C as specified by the procuring activity (see paragraph 6.2). Reference MIL-STD-794 for definition of levels of packing.

*5.2.1 Level A. Transmitters packaged as specified in paragraph 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class. Insofar as practical, exterior shipping containers shall be of uniform shape, size, of minimum tare and cube, consistant with protection required.

*5.2.2 Level B. Transmitter packaged as specified in paragraph 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in paragraph 5.2.1 above shall apply.

*5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to the applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

6. NOTES

6.1 Intended use. The type MK-2 pressure ratio transmitter covered by this specification is intended to indicate (in conjunction with a suitable indicator) the pressure ratio thrust output of an aircraft turbo-jet engine.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When sampling plan B tests will not be conducted.

c. Level of packaging and packing desired.

*6.3 First article tests. The Government activity responsible for First Article Tests is Oklahoma City ALC/MMEO, Tinker AFB, Oklahoma 73145.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 71

Project Number:
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